

MODULAR CONTRACTS

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FACTS

Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Real contracts: Written in language and divided into modular clauses

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Evaluated clause by clause

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Evaluated clause by clause $\Rightarrow$ phenomena not in standard model

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Incompleteness: Clauses omit relevant contingencies

Conservatism: High burden of truth to verify clause

QUESTIONS

Why are contracts modular, with clauses evaluated separately?

What explains loopholes, complexity, landmines, incompleteness,...?

THIS PAPER

Model of modular contracts

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Premises: Atomic terms

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Assumption

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E.g.

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$\equiv$ marginals preserved, not joints

E.g. true data: $\mathbf{a}_j = 110, \mathbf{a}_k = 101$; observations: $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 110$

Tradeoff: Wide clauses: Susceptible to shuffling; Narrow: Neglect joint info

RESULTS

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Narrow clauses

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Extension: High thresholds

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Extension: High thresholds $\implies$ conservatism

MODEL

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(Atomic) Premises

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Clause

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Target policy

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Target policy: $\mathcal{T} \subset \mathcal{A}$ evaluated on true data

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Objective

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Objective: Minimize $c_I \Pr[\llbracket \Phi \rrbracket > \llbracket \mathcal{T} \rrbracket] + c_{II} \Pr[\llbracket \Phi \rrbracket < \llbracket \mathcal{T} \rrbracket]$

MODEL: EXAMPLE

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Premises

MODEL: EXAMPLE

Premises: $\mathbf{a}_{\text{IC}} = \text{“interest coverage } \geq \kappa_{\text{IC}}\text{”}$; $\mathbf{a}_{\text{LR}} = \text{“leverage ratio } \leq \kappa_{\text{LR}}\text{”}$

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(Possible) clauses

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Data: $\mathbf{a}_j^i = 1$ iff premise j satisfied in (unobserved) state i

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Shuffled data

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Shuffled data: Know likelihoods premises hold, not state-by-state match

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Shuffled data: Know likelihoods premises hold, not state-by-state match

Evaluation: $\llbracket \mathcal{M}_m \rrbracket = 1$ if holds with sufficient likelihood

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Target policy: Both IC and LR hold (jointly)

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(Possible) clauses: $\mathcal{M}_{\text{IC}} = \{\mathbf{a}_{\text{IC}}\}$, $\mathcal{M}_{\text{LR}} = \{\mathbf{a}_{\text{LR}}\}$, $\mathcal{M}_{\text{joint}} = \{\mathbf{a}_{\text{IC}}, \mathbf{a}_{\text{LR}}\}$

Contract: $\Phi^{\mathcal{A}} = \{\mathcal{M}_{\text{IC}}, \mathcal{M}_{\text{LR}}\}$ or $\Phi^{\mathcal{T}} = \{\mathcal{M}_{\text{joint}}\}$

Data: $\mathbf{a}_j^i = 1$ iff premise j satisfied in (unobserved) state i

Shuffled data: Know likelihoods premises hold, not state-by-state match

Evaluation: $\llbracket \mathcal{M}_m \rrbracket = 1$ if holds with sufficient likelihood

Target policy: Both IC and LR hold (jointly)

Cost

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Cost: Deem contract holds when target violated and vice versa

DEFINITIONS: PREMISE- AND TARGET-BASED

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Premise-based contract: $\Phi^{\mathcal{A}} := \{\{\mathbf{a}_j\}\}_{\mathbf{a}_j \in \mathcal{T}}$

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Premise-based contract: $\Phi^{\mathcal{A}} := \{\{\mathbf{a}_j\}\}_{\mathbf{a}_j \in \mathcal{T}}$

Target-based contract: $\Phi^{\mathcal{T}} := \{\mathcal{T}\}$

BENCHMARKS

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$\Rightarrow \llbracket \Phi^{\mathcal{A}} \rrbracket = \llbracket \Phi^{\mathcal{T}} \rrbracket$: premise- and target-based coincide

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$\Rightarrow \llbracket \Phi^{\mathcal{A}} \rrbracket = \llbracket \Phi^{\mathcal{T}} \rrbracket$: premise- and target-based coincide

Evaluation reduces to logic; grouping of premises irrelevant

BENCHMARK: $e = 0$

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No shuffling, so $\hat{\mathbf{a}}_j = \mathbf{a}_j$ for all j

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$\Rightarrow$ target-based contract implements target policy: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \mathcal{T} \rrbracket$

LOOPHOLES

DEFINITION: LOOPHOLE STATES

Loophole states LH: Data realizations s.t. $\llbracket \Phi^{\mathcal{A}} \rrbracket \neq \llbracket \Phi^{\mathcal{T}} \rrbracket$ even if $e = 0$

EXAMPLE: LOOPHOLE

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Two premises with three data each and $\tau = 2$:

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Two premises with three data each and $\tau = 2$:

i
1
2
3

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$
1	1
2	1
3	0

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$
1	1
2	1
3	0
# of 1s $\geq \tau$	

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$
1	1
2	1
3	0
# of 1s $\geq \tau$	1

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$
1	1	1
2	1	0
3	0	1
# of 1s $\geq \tau$	1	

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Two premises with three data each and $\tau = 2$:

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# of 1s $\geq \tau$	1	1

Premise-based: $\llbracket \Phi^{\mathcal{A}} \rrbracket$

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Two premises with three data each and $\tau = 2$:

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1	1	1
2	1	0
3	0	1
# of 1s $\geq \tau$	1	1

Premise-based: $\llbracket \Phi^{\mathcal{A}} \rrbracket = \llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$
1	1	1
2	1	0
3	0	1
# of 1s $\geq \tau$	1	1

Premise-based: $[\Phi^A] = [\mathbf{a}_j] \wedge [\mathbf{a}_k] = [110] \wedge [101]$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$
1	1	1
2	1	0
3	0	1
# of 1s $\geq \tau$	1	1

Premise-based: $\llbracket \Phi^A \rrbracket = \llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket \wedge \llbracket 101 \rrbracket = 1 \wedge 1 = 1$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
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# of 1s $\geq \tau$	1	1	

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Target-based:

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
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Premise-based: $\llbracket \Phi^A \rrbracket = \llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket \wedge \llbracket 101 \rrbracket = 1 \wedge 1 = 1$

Target-based: $\llbracket \Phi^T \rrbracket = \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

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Target-based: $[\Phi^T] = [\mathbf{a}_j \wedge \mathbf{a}_k] = [110 \wedge 101]$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

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1	1	1	1
2	1	0	0
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Premise-based: $[\Phi^A] = [\mathbf{a}_j] \wedge [\mathbf{a}_k] = [110] \wedge [101] = 1 \wedge 1 = 1$

Target-based: $[\Phi^T] = [\mathbf{a}_j \wedge \mathbf{a}_k] = [110 \wedge 101] = [100]$

EXAMPLE: LOOPHOLE

Two premises with three data each and $\tau = 2$:

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
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Target-based: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \wedge 101 \rrbracket = \llbracket 100 \rrbracket = 0$

Loophole: Evaluation & conjunction don't commute: $\llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket \neq \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

LOOPHOLE: CONCRETELY

“Leverage ratio $\leq \kappa_{\text{LR}}$ ” holds in 2/3 states

“Interest coverage $\geq \kappa_{\text{IC}}$ ” holds in 2/3 states

Loophole if $\mathbf{a}_{\text{LR}} = 110$ and $\mathbf{a}_{\text{IC}} = 101$:

High assets (LR holds) & low earnings (IC violated) if spend on R&D

Low assets (LR violated) & high earnings (IC holds) if future earnings low

$\Rightarrow \mathbf{a}_{\text{LR}}$ and $\mathbf{a}_{\text{IC}}$ both hold in 1/3 states

Deem that each term holds but one does not

LOOPHOLE: CONCRETELY

“Interest coverage $\geq \kappa_{\text{IC}}$ ” $\wedge$ “Leverage ratio $\leq \kappa_{\text{LR}}$ ” each hold in 2/3 states

Loophole if:

Earnings low when spending on R&D (IC violated) but long term assets okay (LR holds)

Assets low when future earnings low (LR violated) but current earnings okay (IR holds)

“Interest coverage $\geq \kappa_{\text{IC}}$ $\wedge$ “Leverage ratio $\leq \kappa_{\text{LR}}$ ” holds in 1/3 states

Deem that each term holds but one does not hold

SHUFFLE RISK

DEFINITION: SHUFFLE RISK STATES

Shuffle risk states SR: Data realizations s.t. $0 < \Pr\left[\llbracket \Phi^T \rrbracket = 1 \mid \{\mathbf{a}_j\}_j\right] < 1$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
3	0	1	0
$\# \text{ of } 1\text{s} \geq \tau$	1	1	0

EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$
1	1	1	1	1
2	1	0	0	1
3	0	1	0	0
# of 1s $\geq \tau$	1	1	0	1

EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$	$\hat{\mathbf{a}}_k$
1	1	1	1	1	1
2	1	0	0	1	1
3	0	1	0	0	0
# of 1s $\geq \tau$	1	1	0	1	1

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Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$	$\hat{\mathbf{a}}_k$	$\hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k$
1	1	1	1	1	1	1
2	1	0	0	1	1	1
3	0	1	0	0	0	0
# of 1s $\geq \tau$	1	1	0	1	1	1

Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

Joint changed: $101 \rightarrow 110 \Rightarrow$ target-based evaluation flips

SHUFFLE RISK: CONCRETELY

Back to leverage ratio and interest coverage

True data: $\mathbf{a}_{\text{LR}} = 110$ and $\mathbf{a}_{\text{IC}} = 101$:

High assets go with low earnings

Observations = shuffled data: E.g. $\hat{\mathbf{a}}_{\text{LR}} = 110$ and $\hat{\mathbf{a}}_{\text{IC}} = 110$

Serial correlation: High earnings today $\equiv$ high assets

$\Rightarrow \mathbf{a}_{\text{LR}}$ and $\mathbf{a}_{\text{IC}}$ both hold in 2/3 states

RESULTS

LEMMA

LEMMA: LOOPHOLES ARE FALSE POSITIVES

If $[[\Phi^{\mathcal{A}}]] \neq [[\mathcal{T}]]$, then $[[\Phi^{\mathcal{A}}]] = 1$

LEMMA: LOOPHOLES FALSE POSITIVES: PROOF

Suppose (for contradiction) that $\llbracket \mathcal{T} \rrbracket = 1$ and $\llbracket \Phi^{\mathcal{A}} \rrbracket = 0$

$$\llbracket \mathcal{T} \rrbracket = 1 \Rightarrow \bigwedge_{a_j} a_j^i = 1 \text{ for at least } \tau \text{ "rows"}$$

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$$\Rightarrow a_j^i = 1 \text{ for at least } \tau \text{ rows for each } j$$

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$$\Rightarrow a_j^i = 1 \text{ for at least } \tau \text{ rows for each } j$$

$$\Rightarrow [\mathbf{a}_j] = 1 \text{ for each } j$$

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Suppose (for contradiction) that $\llbracket \mathcal{T} \rrbracket = 1$ and $\llbracket \Phi^{\mathcal{A}} \rrbracket = 0$

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$$\Rightarrow \llbracket \mathbf{a}_j \rrbracket = 1 \text{ for each } j$$

$$\Rightarrow \llbracket \Phi^{\mathcal{A}} \rrbracket = 1$$

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$$\Rightarrow a_j^i = 1 \text{ for at least } \tau \text{ rows for each } j$$

$$\Rightarrow \llbracket \mathbf{a}_j \rrbracket = 1 \text{ for each } j$$

$$\Rightarrow \llbracket \Phi^{\mathcal{A}} \rrbracket = 1$$

$$\Rightarrow \Leftarrow$$

LEMMA

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK

LH $\subsetneq$ SR

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR: Loophole states: Premise-based accepts, target-based rejects

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR: Loophole states: Premise-based accepts, target-based rejects

Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

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SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$

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SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $\llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket = 1$ if $\tau = 2$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR: Loophole states: Premise-based accepts, target-based rejects

Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

Target-based accepts too if shuffle “aligns” all 1’s

SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $\llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket = 1$ if $\tau = 2$ (not LH)

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

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Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

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But shuffle could be $\hat{\mathbf{a}}_j = 110$, $\hat{\mathbf{a}}_k = 101$

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SR $\nRightarrow$ LH: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $\llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \rrbracket = 1$ if $\tau = 2$ (not LH)

But shuffle could be $\hat{\mathbf{a}}_j = 110$, $\hat{\mathbf{a}}_k = 101$: $\llbracket \hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k \rrbracket = \llbracket 100 \rrbracket = 0$ (yes SR)

R1: LOOPHOLES

R1: LOOPHOLES

With $e > 0$, if $\Pr[\text{SR} \setminus \text{LH}]$ large, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

Shuffled data ($e > 0$): Evaluation errors for all but not only loopholes

Accept loopholes to reduce shuffling error

LOOPHOLES IN PRACTICE: SERTA

Serta was distressed and wanted to raise liquidity by priming lenders

Credit agreement ostensibly prohibited non-pro-rata treatment, but...

Clause $\mathbf{a}_j$: “Majority can amend the credit agreement”

Clause $\mathbf{a}_k$: “Amendment can permit open market purchase”

Clause $\mathbf{a}_\ell$: “Selective exchange = open market purchase”

LOOPHOLES IN PRACTICE: SERTA

Serta was distressed and wanted to raise liquidity by priming lenders

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Clause $\mathbf{a}_j$: “Majority can amend the credit agreement”

Clause $\mathbf{a}_k$: “Amendment can permit open market purchase”

Clause $\mathbf{a}_\ell$: “Selective exchange = open market purchase”

$\Rightarrow \mathbf{a}_j \wedge \mathbf{a}_k \wedge \mathbf{a}_\ell$: Can prime non-participating lenders via exchange

(Exchange made participating lenders senior, pushing others down)

LOOPHOLES IN PRACTICE: SERTA

Serta was distressed and wanted to raise liquidity by priming lenders

Credit agreement ostensibly prohibited non-pro-rata treatment, but...

Clause $\mathbf{a}_j$: “Majority can amend the credit agreement”

Clause $\mathbf{a}_k$: “Amendment can permit open market purchase”

Clause $\mathbf{a}_\ell$: “Selective exchange = open market purchase”

$\Rightarrow \mathbf{a}_j \wedge \mathbf{a}_k \wedge \mathbf{a}_\ell$: Can prime non-participating lenders via exchange

(Exchange made participating lenders senior, pushing others down)

I.e. loophole: $\llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket \wedge \llbracket \mathbf{a}_\ell \rrbracket \neq \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \wedge \mathbf{a}_\ell \rrbracket$

LOOPHOLES IN THE LITERATURE

Katz 10: Many loopholes openly exploited after much litigation

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Show why loopholes persist $e > 0$

R2: ERRORS DUE TO COMPLEXITY

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If $c_I \text{Pr}[\text{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

R2: COMPLEXITY RISK: INTUITION

If loopholes likely or costly use complex target-based contract (long clauses)

Cost: With interacting premises, shuffle risk $\Rightarrow$ likely evaluated with error

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If loopholes likely or costly use complex target-based contract (long clauses)

Cost: With interacting premises, shuffle risk $\Rightarrow$ likely evaluated with error

Empirically: Complexity leads to errors in taxes and regulatory compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if $a_j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for some clauses $a_j \in \mathcal{A}$, $m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_1 \text{Pr}[\text{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

Benefit w.r.t. premise-based: Fewer loopholes

Keeps joint information

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

Benefit w.r.t. premise-based: Fewer loopholes

Keeps joint information

Cost: False negatives: Shuffles in any clause can cause rejection

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$\Phi^{\mathcal{I}}$ called incomplete w.r.t. $\mathcal{T}$ if $j \notin \bigcup \mathcal{M}_m$ for some $j \in \mathcal{T}$

R4: INCOMPLETENESS

If e and c_{Π} high and $\Pr [\llbracket \Phi^{\mathcal{I}} \rrbracket > \llbracket \phi^{\mathcal{T}} \rrbracket]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS: INTUITION

Long clauses induce shuffling risk $\Rightarrow$ omit some premises to avoid it

Akin to mismeasurement in Holmström–Milgrom:

Decrease power on distortionary tasks

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is conservative if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM

Define $\text{ACC}_\tau(\Phi) := \{ \{ \mathbf{a}_j \}_j : \llbracket \Phi \rrbracket = 1 \}$

If $\Pr [\llbracket \mathcal{T} \rrbracket = 1 \mid \text{ACC}_\tau(\Phi^{\mathcal{A}}) \setminus \text{ACC}_{\tau+1}(\Phi^{\mathcal{A}})] < \frac{c_{\text{I}}}{c_{\text{I}} + c_{\text{II}}}$, then $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{A}}$

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

R5: CONSERVATISM: APPLICATIONS

Law: “beyond reasonable doubt,” “reasonable suspicion,” “probable cause”
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

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Model of modular contracts in formal language s.t. shuffled data

Loopholes

Errors of complexity

Landmines

Incompleteness

Conservatism

MODULAR CONTRACTS